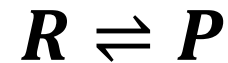


Chapter 15



Chemical Equilibrium

- ▶ The Concept of Equilibrium
- ▶ The Equilibrium Constant
- ▶ Understanding and Working with Equilibrium Constants
- ▶ Calculating Equilibrium Constants
- ▶ Le Châtelier's Principle

The Concept of Equilibrium

Q. What is a reversible process (reaction)?



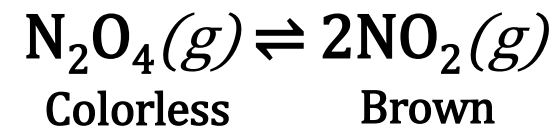
Colorless Gas
(Reactant)

Brown Gas
(Product)

- Most chemical reactions do not go to 100% completion.
 - Instead, reactants convert to products, and products simultaneously reform reactants.
 - **Reversible reaction:** A process in which products can react to regenerate the original reactants.
 - This bidirectional process is represented in chemical equations by a double arrow ($\rightleftharpoons$).
- **Chemical equilibrium** occurs when a reaction and its reverse reaction proceed at the same rate.

The Concept of Equilibrium

Q. What happens when a reaction reaches equilibrium?



Initial State

Starting with pure N_2O_4 :

- The solution is entirely colorless.
- Forward reaction rate is at its maximum because reactant concentration is highest.
- Reverse reaction rate is zero (no NO_2 is present).

Reaction Progresses

As N_2O_4 decomposes to NO_2 :

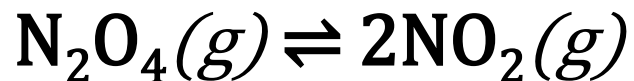
- Brown color begins to appear and intensify.
- Concentration of N_2O_4 decreases, slowing down the forward rate.
- Concentration of NO_2 increases, accelerating the reverse rate.

Equilibrium Achieved

At a certain point (in a **closed system**):

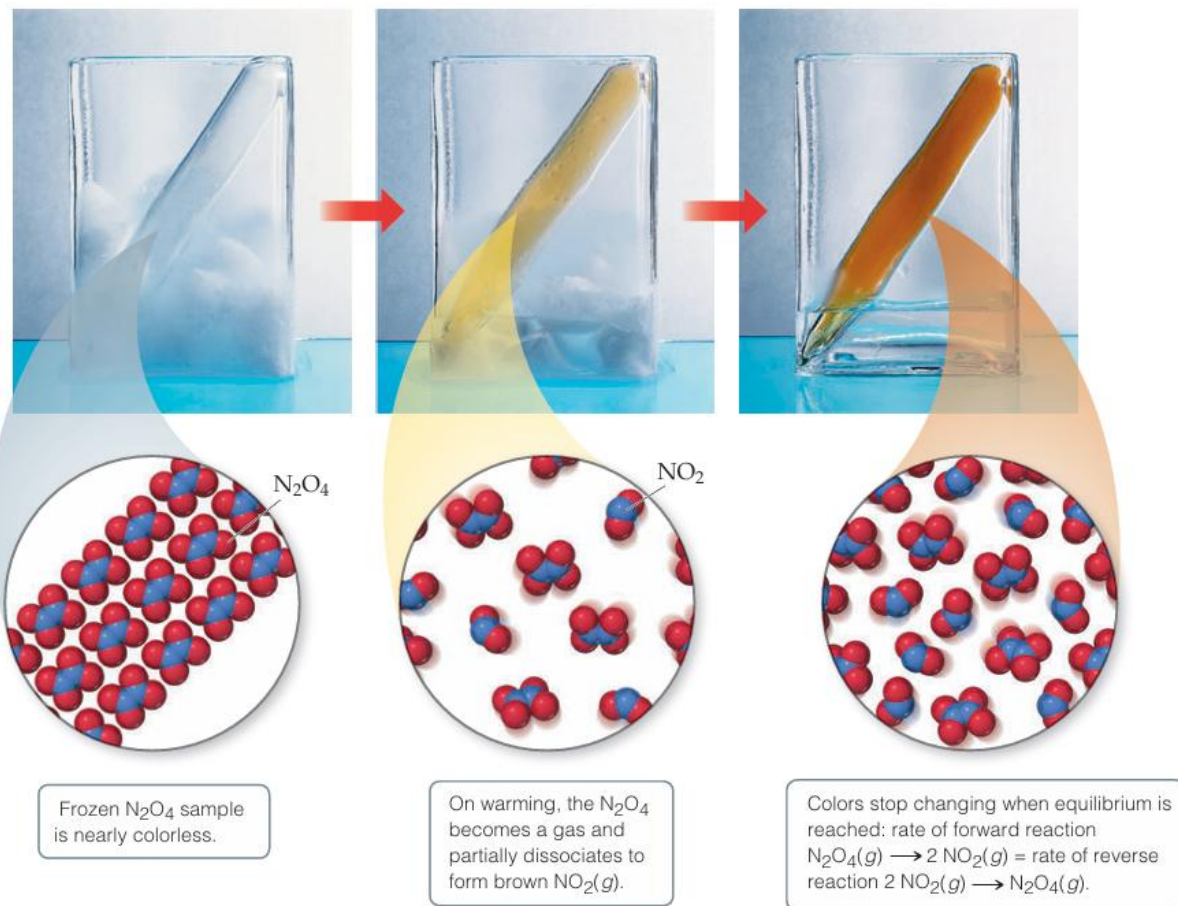
- $[\text{N}_2\text{O}_4]$ and $[\text{NO}_2]$ no longer change with time.
- **Rates of forward = Rate of reverse.**
- Both reactions continue, but a particular ratio of concentration terms equals a constant.

The Concept of Equilibrium

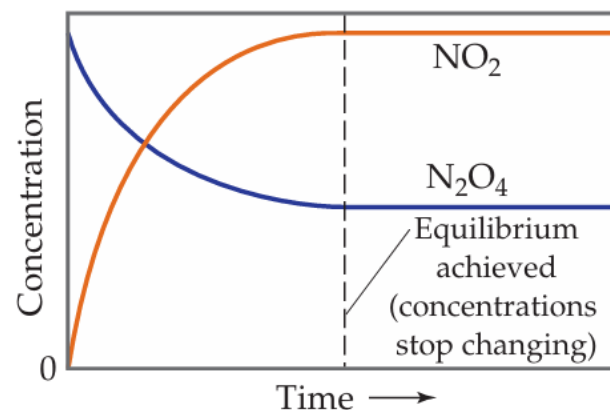


Colorless

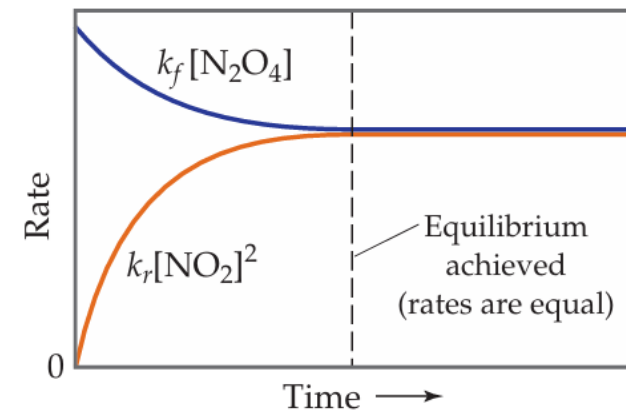
Brown



The equilibrium between NO_2 and N_2O_4 .



(a)

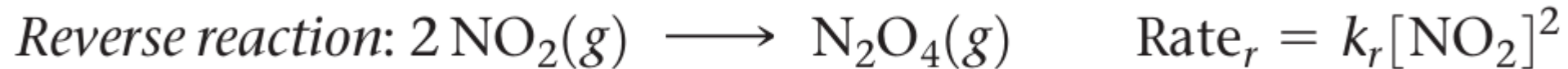
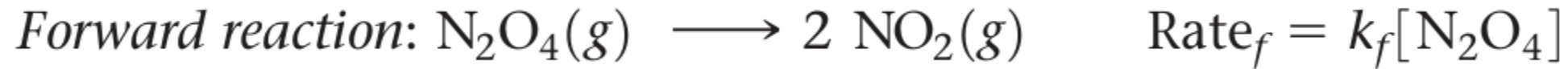
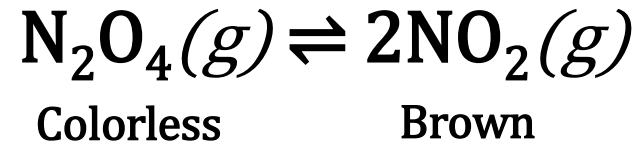


(b)

Achieving chemical equilibrium. (Equilibrium occurs when the rate of the forward reaction equals the rate of the reverse reaction).

Q. Is the concentration of reactants and products equal at equilibrium?

The Concept of Equilibrium



At equilibrium:

$$\underset{\text{Forward reaction}}{k_f[\text{N}_2\text{O}_4]} = \underset{\text{Reverse reaction}}{k_r[\text{NO}_2]^2}$$

$$\frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = \frac{k_f}{k_r} = \text{a constant}$$

- The quotient of two rate constants is another constant (**the equilibrium constant**)

The Equilibrium Constant

Q. How can we determine the relationship between the concentrations of reactants and products at **equilibrium**?

- **The law of mass action (1864):**
Expresses the relationship between the concentrations of the reactants and products present at equilibrium
- Developed by Cato Maximilian Guldberg (1836–1902) and Peter Waage (1833–1900).
- ✓ The equilibrium condition is described by “**equilibrium-constant expression**” (equilibrium expression).

Consider a general reaction:

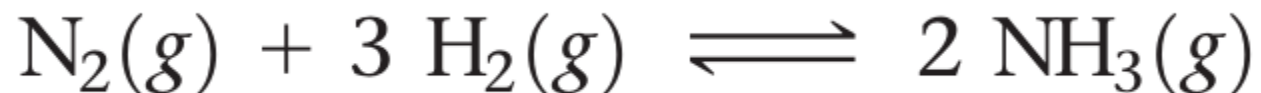


$$K_c = \frac{[D]^d [E]^e}{[A]^a [B]^b} \quad \begin{array}{l} \leftarrow \text{products} \\ \leftarrow \text{reactants} \end{array}$$

- ✓ K_c : The equilibrium constant.
- ✓ a,b,c,d: stoichiometric coefficients
- ✓ The equilibrium constant depends on stoichiometry.

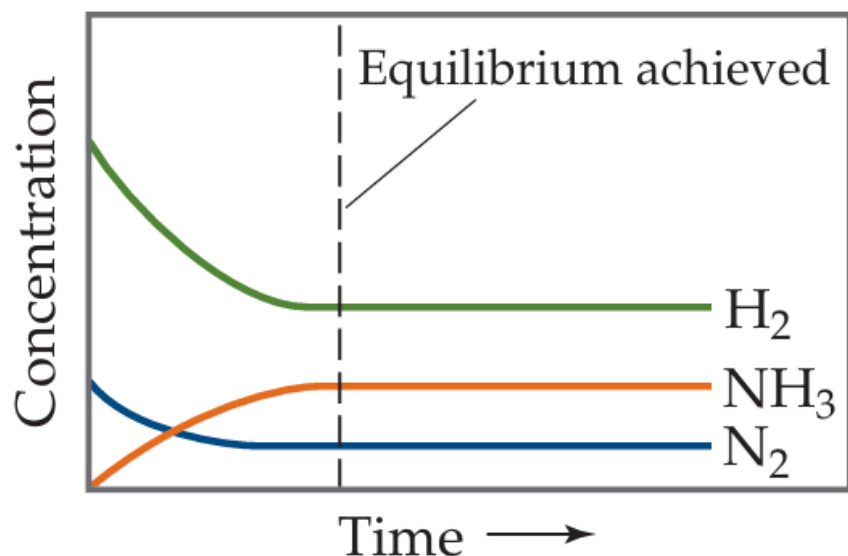
The Equilibrium Constant

Example: The Haber Process

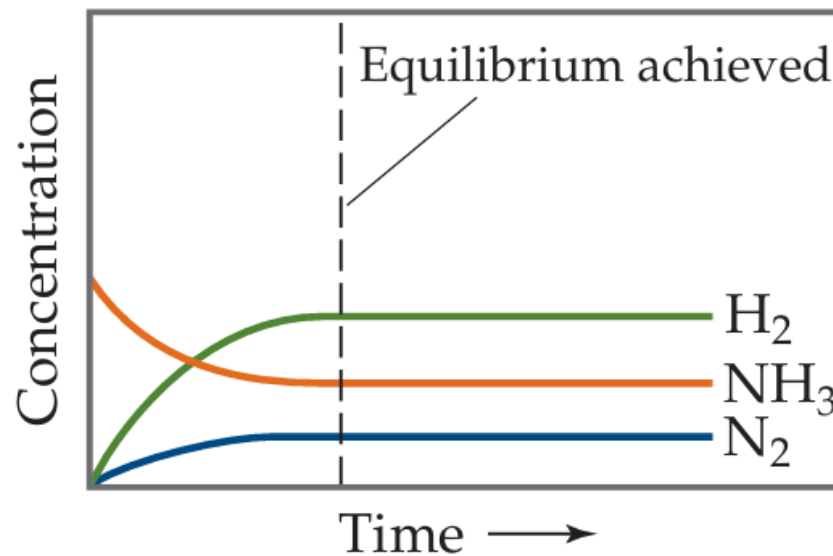


$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

Starting with only reactants



Starting with only products



The same equilibrium is reached whether we start with only reactants or with only product.



The Nobel Prize in Chemistry (1918)

The Equilibrium Constant

Q. How do we evaluate K_c from the law of mass action?

Example:

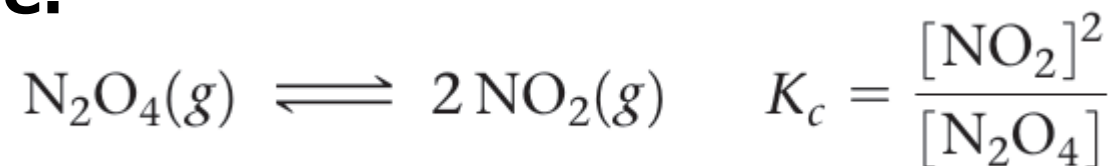
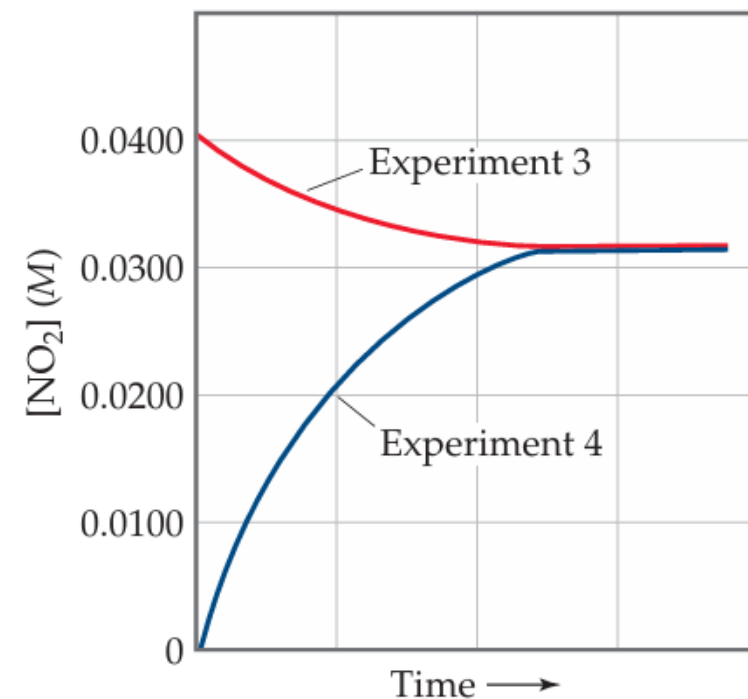


TABLE 15.1 Initial and Equilibrium Concentrations of $\text{N}_2\text{O}_4(g)$ and $\text{NO}_2(g)$ at 100 °C

Experiment	Initial $[\text{N}_2\text{O}_4] (M)$	Initial $[\text{NO}_2] (M)$	Equilibrium $[\text{N}_2\text{O}_4] (M)$	Equilibrium $[\text{NO}_2] (M)$	K_c
1	0.0	0.0200	0.00140	0.0172	0.211
2	0.0	0.0300	0.00280	0.0243	0.211
3	0.0	0.0400	0.00452	0.0310	0.213
4	0.0200	0.0	0.00452	0.0310	0.213

$$K_c = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = \frac{[0.0172]^2}{0.00140} = 0.211 \quad (\text{using Experiment 1 data})$$

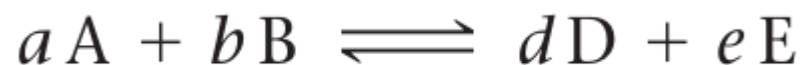


The Equilibrium Constant

Q. How do we evaluate K_c from the law of mass action?

Equilibrium Constants in Terms of Pressure, K_p

- **Gas-phase equilibrium:** Equilibrium expressions can be written using the partial pressures of gaseous reactants and products.
- **Equilibrium constant (K_p):** When partial pressures are measured in atmospheres (atm), the equilibrium constant is called K_p ("p" = pressure).



$$K_p = \frac{(P_D)^d (P_E)^e}{(P_A)^a (P_B)^b}$$

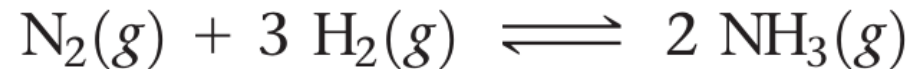
P : partial pressure



$$K_p = \frac{(P_{\text{NO}_2})^2}{P_{\text{N}_2\text{O}_4}}$$

The Equilibrium Constant

- Q. After a mixture of hydrogen and nitrogen gases in a reaction vessel is allowed to attain equilibrium at 472 °C, it is found to contain 7.38 bar H₂, 2.46 bar N₂, and 0.166 bar NH₃. From these data, calculate the equilibrium constant K_p for the reaction.



$$K_p = \frac{(P_{\text{NH}_3})^2}{P_{\text{N}_2}(P_{\text{H}_2})^3} = \frac{0.166^2}{(2.46)(7.38)^3} = 2.71 \times 10^{-9}$$

- Q. Is it possible to determine K_c for the above reaction?

The Equilibrium Constant

Correlation between K_c and K_p



$$K_p = \frac{(P_D)^d (P_E)^e}{(P_A)^a (P_B)^b}$$

$$K_c = \frac{[D]^d [E]^e}{[A]^a [B]^b}$$

Q. What are the units of equilibrium constants?

$$PV = nRT, \text{ so } P = \frac{n}{V}RT$$

$$P_A = \frac{n_A}{V}RT = [A]RT$$

- It is possible to calculate one from the other using the **ideal-gas equation**.
- $n/V = \text{mol/L}$ (equals molarity, M)
- For substance A in our generic reaction

$$K_p = \frac{([D]RT)^d ([E]RT)^e}{([A]RT)^a ([B]RT)^b} = \left(\frac{[D]^d [E]^e}{[A]^a [B]^b} \right) \frac{(RT)^{d+e}}{(RT)^{a+b}} = K_c (RT)^{(d+e)-(a+b)} = K_c (RT)^{\Delta n}$$

The Equilibrium Constant

Correlation between K_c and K_p



$$K_p = K_c(RT)^{\Delta n}$$

- **R:** Universal gas constant = 0.08206 *L·atm / mol·K*, 8.314 *L.kPa/ mol.K*)
- **T:** Absolute temperature in Kelvin (K)
- **Δn :** Moles of gaseous product - Moles of gaseous reactant

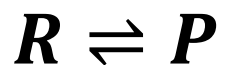
Q. For the Haber process, K_c is 9.60 at 300 °C. Calculate K_p for this reaction at this temperature.



$$K_p = K_c (RT)^{\Delta n} = (9.60)(8.314 \times 573)^{-2} = \frac{(9.60)}{(8.314 \times 573)^2} = 4.23 \times 10^{-10}$$

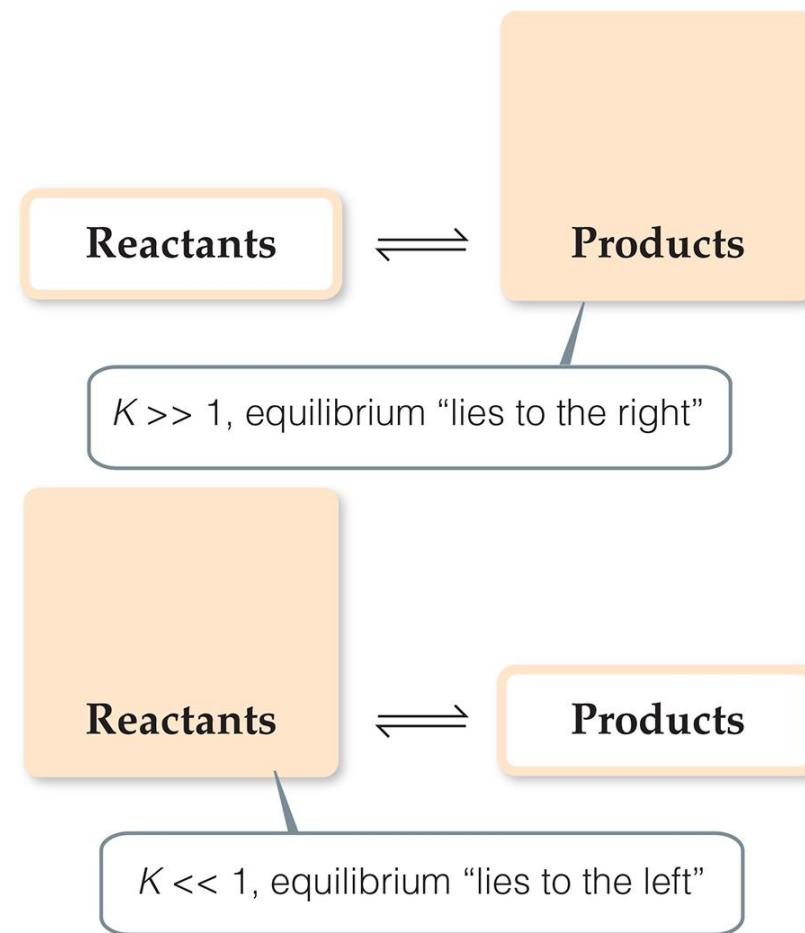
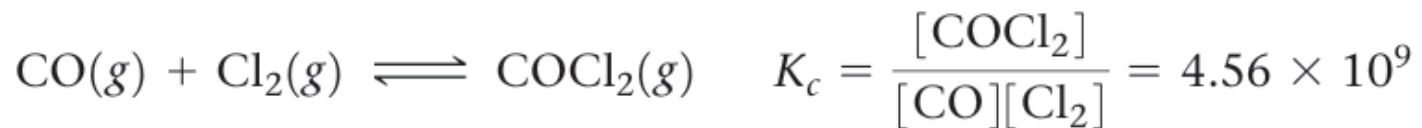
Understanding and Working with Equilibrium Constants

Q. What information can be obtained from the magnitude of equilibrium constants?



- If $K \gg 1$ (large K): the equilibrium lies to right; products predominate.
- If $K \ll 1$ (small K): the equilibrium lies to left; reactants predominate.

Q. Explain the nature of the following reaction at equilibrium.



Relationship between magnitude of K and composition of an equilibrium mixture.

Understanding and Working with Equilibrium Constants

Rules for Manipulating Equilibrium Constants



$$K_c = \frac{[D]^d [E]^e}{[A]^a [B]^b}$$

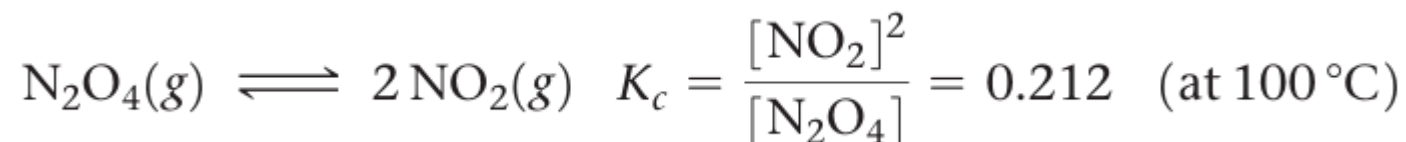
- Q. What will be the K_c for the reverse process under the same reaction conditions?
- Q. What will happen to the K_c value if the coefficients are multiplied by a factor of n ?
- Q. What will be the value of K_c if the reaction occurs in multistep processes?

Understanding and Working with Equilibrium Constants

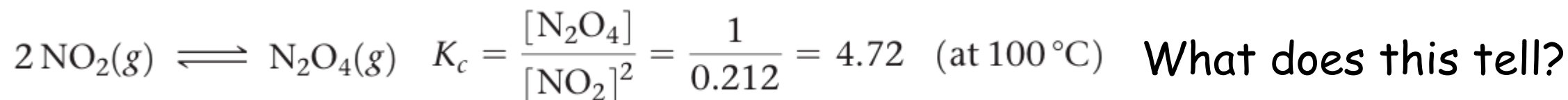
Rules for Manipulating Equilibrium Constants

The Direction of the Chemical Equation and K

Consider:



- We could equally well consider this equilibrium in terms of the reverse reaction:



- ✓ The equilibrium expression for a reaction written in one direction is the reciprocal of the expression for the reaction written in the reverse direction.

Understanding and Working with Equilibrium Constants

Rules for Manipulating Equilibrium Constants

Stoichiometry and Equilibrium Constants

Consider: $\text{N}_2\text{O}_4(g) \rightleftharpoons 2 \text{NO}_2(g)$

$$K_c = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = 0.212 \quad (\text{at } 100^\circ\text{C})$$

- If we multiply the reaction equation by 2:



$$K_c = \frac{[\text{NO}_2]^4}{[\text{N}_2\text{O}_4]^2}$$

$$K_c = 0.212^2 = 0.0449 \quad (\text{at } 100^\circ\text{C})$$

✓ The concentrations of the substances in the equilibrium mixture will be the same no matter how you write the chemical equation, but the value of K_c you calculate depends on how you write the reaction.

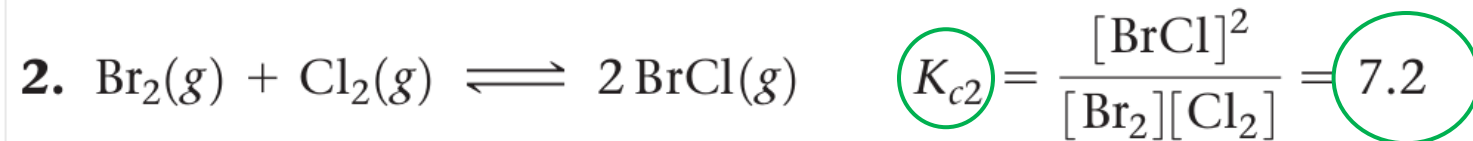
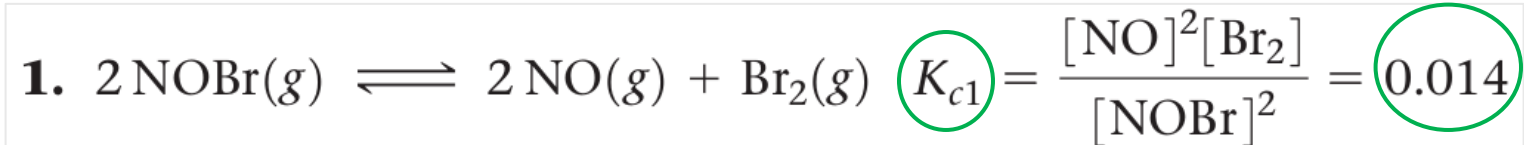
What does this tell?

Understanding and Working with Equilibrium Constants

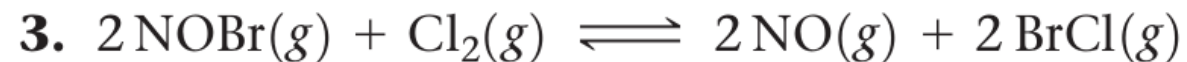
Rules for Manipulating Equilibrium Constants

Equilibrium constant for a multistep reaction

- Consider the following two reactions at 100 °C:



- The net sum of these two equations is:



$$K_{c3} = \frac{[\text{NO}]^2[\text{BrCl}]^2}{[\text{NOBr}]^2[\text{Cl}_2]} = \frac{[\text{NO}]^2[\text{Br}_2]}{[\text{NOBr}]^2} \times \frac{[\text{BrCl}]^2}{[\text{Br}_2][\text{Cl}_2]}$$

$$K_{c3} = (K_{c1})(K_{c2}) = (0.014)(7.2) = 0.10$$

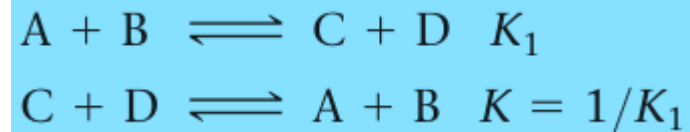
✓ The equilibrium expression for the net reaction is the product of the expressions for individual reactions.

What does this tell?

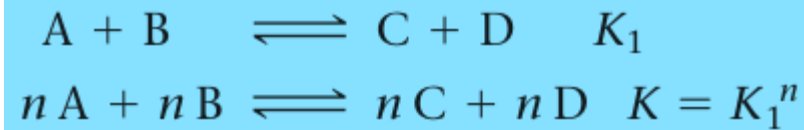
Understanding and Working with Equilibrium Constants

To summarize:

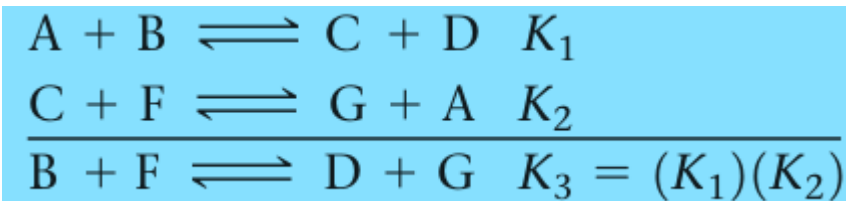
1. The equilibrium constant of a reaction in the reverse direction is the inverse (or reciprocal) of the equilibrium constant of the reaction in the forward direction:



2. The equilibrium constant of a reaction that has been multiplied by a number is equal to the original equilibrium constant raised to a power equal to that number:

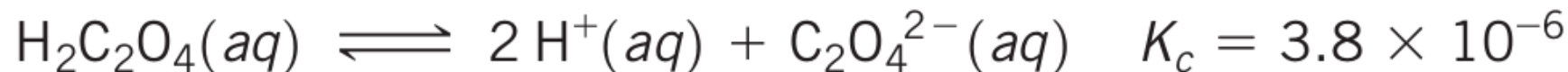
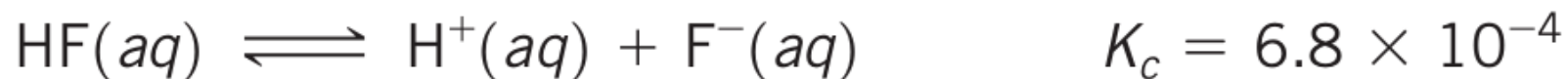


3. The equilibrium constant for a net reaction made up by adding two or more reactions is the product of the equilibrium constants for the individual reactions



Understanding and Working with Equilibrium Constants

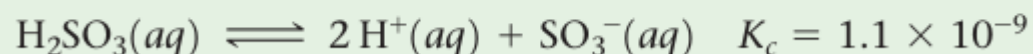
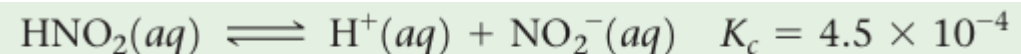
Q. Given the reactions:



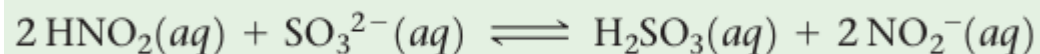
determine the value of K_c for the reaction:



Q. Given the equilibrium constants for the following two reactions in aqueous solution at 25 °C:



calculate is the value of K_c for the reaction:



Understanding and Working with Equilibrium Constants

Homogeneous Vs Heterogeneous Equilibria

Homogeneous Equilibrium

- All reacting species are in the same phase (either all gases or all dissolved in solution).
- Example reaction (all gases):
 - $2\text{NO}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{NO}_2(g)$
- Every species appears in the K_c expression because their concentrations vary during the reaction.

Heterogeneous Equilibrium

- Reactants and products are present in more than one phase (e.g. solids, liquids, and gases).

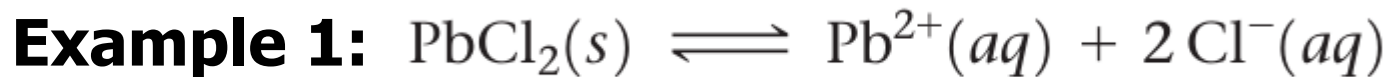


Q. How do we express the equilibrium constant for a heterogeneous equilibrium?

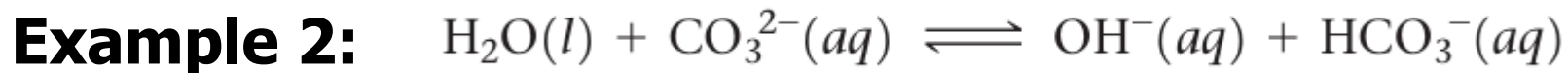
Understanding and Working with Equilibrium Constants

Heterogeneous Equilibria

- Q. How do we express the equilibrium constant for heterogeneous equilibrium?
- Whenever a **pure solid** or a **pure liquid** is involved in a heterogeneous equilibrium, its concentration is not included in the equilibrium expression. Why?



$$K_c = [\text{Pb}^{2+}][\text{Cl}^{-}]^2$$

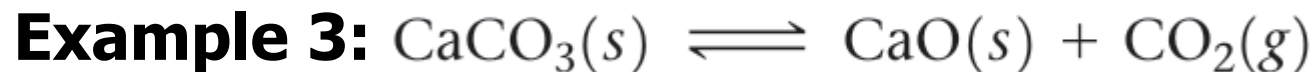


$$K_c = \frac{[\text{OH}^{-}][\text{HCO}_3^{-}]}{[\text{CO}_3^{2-}]}$$

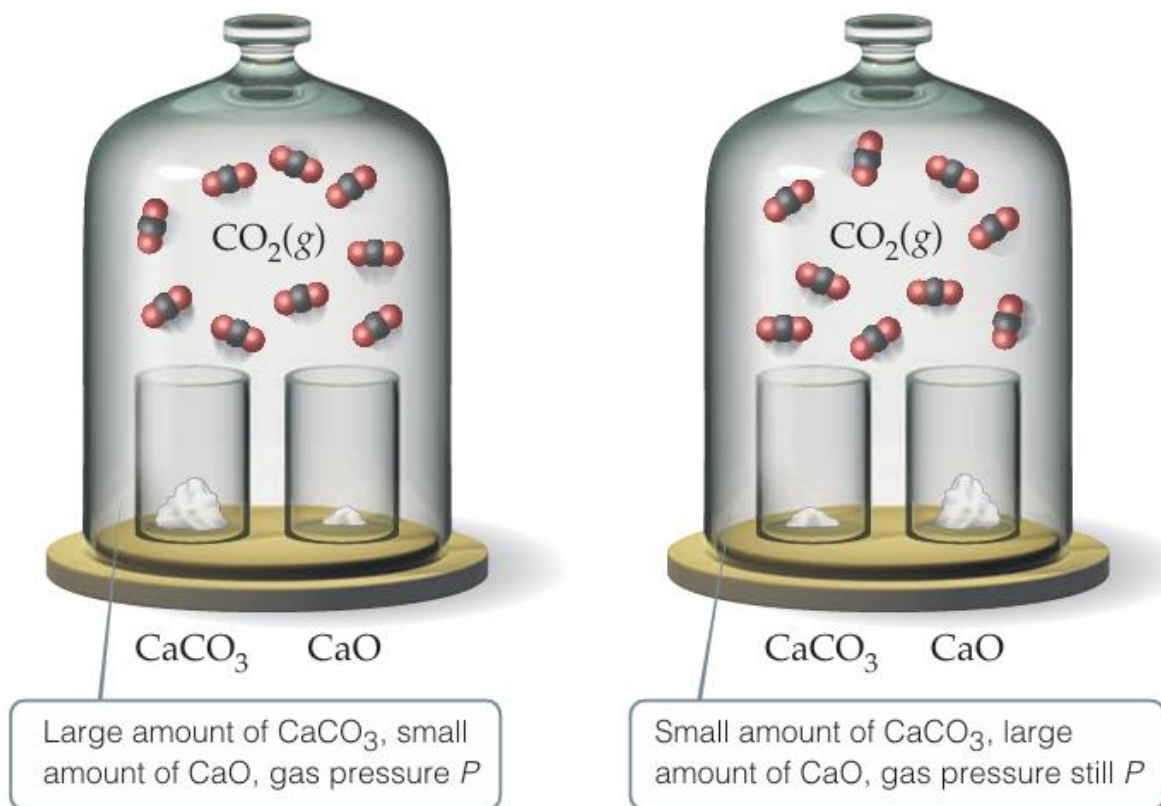
- Q. Why H_2O is not included in the equilibrium constant expression?

Understanding and Working with Equilibrium Constants

Heterogeneous Equilibria

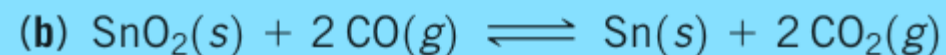
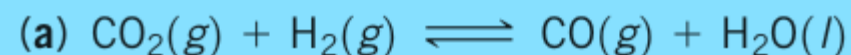


$$K_c = [\text{CO}_2] \quad \text{and} \quad K_p = P_{\text{CO}_2}$$



- At a given temperature, the equilibrium pressure of CO₂ in the bell jars is the same no matter how much of each solid is present.

Write the equilibrium expression K_c for



Calculating Equilibrium Constants

Q. How to Determine Concentrations of Unknown Species in an Equilibrium Mixture?

1. Tabulate all known initial and equilibrium concentrations of the species that appear in the equilibrium expression.
2. For those species for which initial and equilibrium concentrations are known, calculate the change in concentration that occurs as the system reaches equilibrium.
3. Use the stoichiometry of the reaction (that is, the coefficients in the balanced chemical equation) to calculate the changes in concentration for all other species in the equilibrium expression.
4. Use initial concentrations from step 1 and changes in concentration from step 3 to calculate any equilibrium concentrations not tabulated in step 1.
5. Determine the value of the equilibrium constant.

Calculating Equilibrium Constants

Q. A reaction vessel containing $1.000 \times 10^{-3} \text{ M}$ H_2 gas and $2.000 \times 10^{-3} \text{ M}$ I_2 gas is heated to 448°C as: $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2 \text{HI}(\text{g})$. What is the value of the equilibrium constant K_c if once the system comes to equilibrium at 448°C and the concentration of HI is $1.87 \times 10^{-3} \text{ M}$?

Condition	$[\text{H}_2], \text{M}$	$[\text{I}_2], \text{M}$	$[\text{HI}], \text{M}$
Initial	1.000×10^{-3}	2.000×10^{-3}	0
Change	-9.35×10^{-4}	-9.35×10^{-4}	$+1.87 \times 10^{-3}$
Equilibrium	6.5×10^{-5}	1.065×10^{-3}	1.87×10^{-3}

1. What Do We Know (initial, given)?

2. $[\text{HI}]$ Increases by $1.87 \times 10^{-3} \text{ M}$ (change)

3. Stoichiometry tells us $[\text{H}_2]$ and $[\text{I}_2]$ decrease by half as much (change)

$$\left(1.87 \times 10^{-3} \frac{\text{mol HI}}{\text{L}}\right) \left(\frac{1 \text{ mol H}_2}{2 \text{ mol HI}}\right) = 0.935 \times 10^{-3} \frac{\text{mol H}_2}{\text{L}}$$

$$\left(1.87 \times 10^{-3} \frac{\text{mol HI}}{\text{L}}\right) \left(\frac{1 \text{ mol I}_2}{2 \text{ mol HI}}\right) = 0.935 \times 10^{-3} \frac{\text{mol I}_2}{\text{L}}$$

4. Calculate the equilibrium concentrations of H_2 and I_2

$$[\text{H}_2] = (1.000 \times 10^{-3} \text{ M}) - (0.935 \times 10^{-3} \text{ M}) = 0.065 \times 10^{-3} \text{ M}$$

$$[\text{I}_2] = (2.000 \times 10^{-3} \text{ M}) - (0.935 \times 10^{-3} \text{ M}) = 1.065 \times 10^{-3} \text{ M}$$

$$5. \quad K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

$$= \frac{(1.87 \times 10^{-3})^2}{(0.065 \times 10^{-3})(1.065 \times 10^{-3})} = 51$$

Applications of Equilibrium Constants



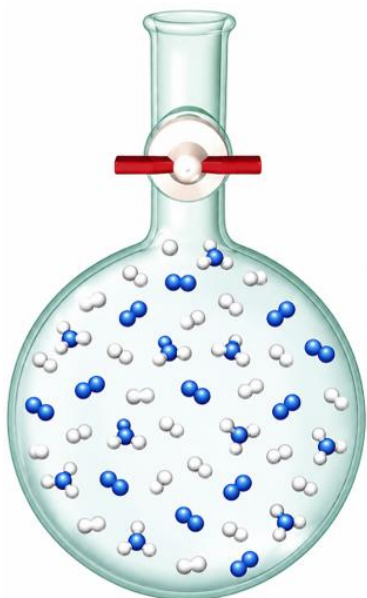
$$K_p = \frac{(P_D)^d (P_E)^e}{(P_A)^a (P_B)^b}$$

$$K_c = \frac{[D]^d [E]^e}{[A]^a [B]^b}$$

- The magnitude of K indicates the extent to which a reaction proceeds.
 - ✓ $K \gg 1 \rightarrow$ mostly products
 - ✓ $K \ll 1 \rightarrow$ mostly reactants
- The equilibrium constant can also be used to:
 - ✓ Predict the direction in which a reaction mixture achieves equilibrium
 - ✓ Calculate equilibrium concentrations of reactants and products.

Predicting the Direction of Reaction

Consider the following condition:



1 mol of N_2

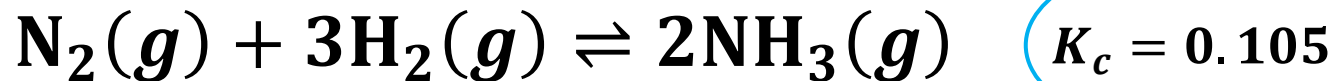
2 mol H_2

2 mol NH_3

Volume of container = 1L

$T = 472\text{ }^{\circ}\text{C}$

(Closed system)



Q. How will the mixture react to reach equilibrium?

Q. Will N_2 and H_2 react to form more NH_3 , or will NH_3 decompose to N_2 and H_2 ?

$$\begin{array}{l} [N_2] = 1.00\text{ M} \\ [H_2] = 2.00\text{ M} \\ [NH_3] = 2.00\text{ M} \end{array} \quad \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(2.00)^2}{(1.00)(2.00)^3} = 0.500$$

- To reach equilibrium, **the quotient** $[NH_3]^2/[N_2][H_2]^3$ must decrease from the starting value of 0.500 to the equilibrium value of 0.105.
- This change can happen only if $[NH_3]$ decreases and $[N_2]$ and $[H_2]$ increase.
- Thus, the reaction proceeds toward equilibrium by forming N_2 and H_2 from NH_3 .

Predicting the Direction of Reaction

- This approach can be formalized by defining a quantity called the **reaction quotient**.

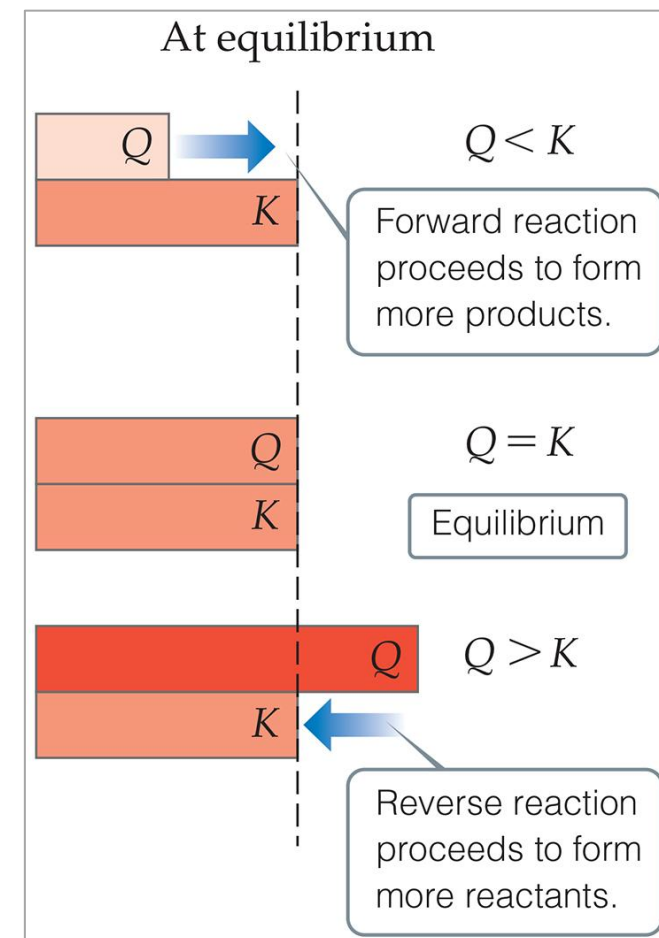
The reaction quotient, Q , is a number obtained by substituting reactant and product concentrations (Q_c) or partial pressures (Q_p) at any point during a reaction into an equilibrium expression.

- For the general reaction: $aA + bB \rightleftharpoons dD + eE$

$$Q_c = \frac{[D]^d[E]^e}{[A]^a[B]^b}$$

How to Use Q to Analyze Reaction Progress?

- $Q < K$: Too few products $\rightarrow$ reaction proceeds left $\rightarrow$ right to form more products.
- $Q = K$: System is at equilibrium.
- $Q > K$: Too many products $\rightarrow$ reaction proceeds right $\rightarrow$ left to form more reactants.



Predicting the direction of a reaction by comparing Q and K at a given temperature.

Predicting the Direction of Reaction

Q. At 448 °C, the equilibrium constant K_c for the following reaction is 50.5. Predict in which direction the reaction proceeds to reach equilibrium if we start with 2.0×10^{-2} mol of HI, 1.0×10^{-2} mol of H_2 , and 3.0×10^{-2} mol of I_2 in a 2.00 L container.



$$Q_c = \frac{[D]^d[E]^e}{[A]^a[B]^b}$$

$$Q_c = \frac{[HI]^2}{[H_2][I_2]}$$

The initial concentrations are

$$[HI] = 2.0 \times 10^{-2} \text{ mol}/2.00 \text{ L} = 1.0 \times 10^{-2} \text{ M}$$

$$[H_2] = 1.0 \times 10^{-2} \text{ mol}/2.00 \text{ L} = 5.0 \times 10^{-3} \text{ M}$$

$$[I_2] = 3.0 \times 10^{-2} \text{ mol}/2.00 \text{ L} = 1.5 \times 10^{-2} \text{ M}$$

$$Q_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(1.0 \times 10^{-2})^2}{(5.0 \times 10^{-3})(1.5 \times 10^{-2})} = 1.3$$

$Q_c < K_c$: The reaction as written proceeds left to right to attain equilibrium.

Calculating Equilibrium Concentrations

- Chemists often need to determine the amounts of reactants and products present at equilibrium when the equilibrium constant (K) is known.
- General Approach: using the ICE Table.

	Reactants	Products
Initial	Initial concentration/pressure	Initial concentration/pressure
Change	$-x$	$+x$
Equilibrium	Initial $- x$	Initial $+ x$

Calculating Equilibrium Concentrations

Q. A 1.000 L flask is filled with 1.000 mol of $\text{H}_2(g)$ and 2.000 mol of $\text{I}_2(g)$ at 448 °C. Given a K_c of 50.5 at 448 °C, what are the equilibrium concentrations of H_2 , I_2 , and HI ?



	H_2	I_2	HI
Initial (M)	1.000	2.000	0
Change (M)	-x	-x	+2x
Equilibrium(M)	1.000 - x	2.000 - x	2x

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(2x)^2}{(1.000 - x)(2.000 - x)} = 50.5$$

$$46.5x^2 - 151.5x + 101.0 = 0 \quad \text{Quadratic equation}$$

$$x = \frac{-(-151.5) \pm \sqrt{(-151.5)^2 - 4(46.5)(101.0)}}{2(46.5)} = 2.323 \text{ or } 0.935$$

$$[\text{H}_2] = 1.000 - x = 0.065 M$$

$$[\text{I}_2] = 2.000 - x = 1.065 M$$

$$[\text{HI}] = 2x = 1.87 M$$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Calculating Equilibrium Concentrations

Q. For the Haber process, $K_p = 1.41 \times 10^{-9}$, at 500 °C. In an equilibrium mixture of the three gases at 500 °C, the partial pressure of H_2 is 94.03 bar and that of N_2 is 43.77 bar. What is the partial pressure of NH_3 in this equilibrium mixture?

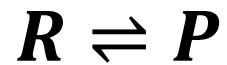


$$K_p = \frac{(P_{NH_3})^2}{P_{N_2}(P_{H_2})^3} = \frac{x^2}{(43.77)(94.03)^3} = 1.41 \times 10^{-9}$$

$$x^2 = (1.41 \times 10^{-9})(43.77)(94.03)^3 = 5.14 \times 10^{-2}$$

$$x = \sqrt{5.14 \times 10^{-2}} = 0.227 \text{ bar} = P_{NH_3}$$

Le Châtelier's Principle (Henri-Louis Le Châtelier (1850–1936))



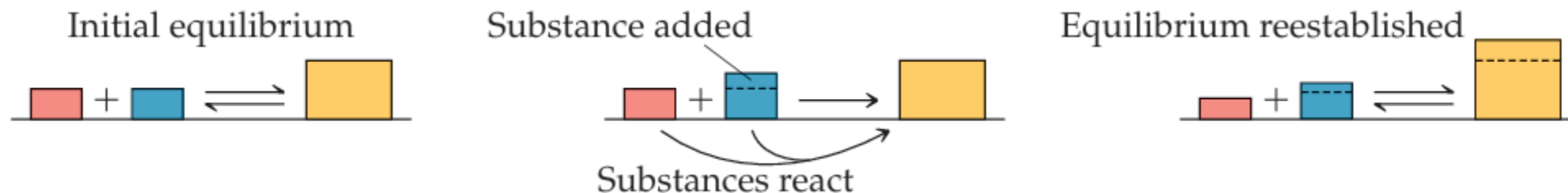
Q. How will a change in conditions (T, P, C) affect the position of equilibrium?

Le Châtelier's Principle:

"If a system at equilibrium is disturbed by a change in temperature, pressure, or a component concentration, the system will shift its equilibrium position so as to counteract the effect of the disturbance."

- **Concentration:** adding or removing a reactant or product

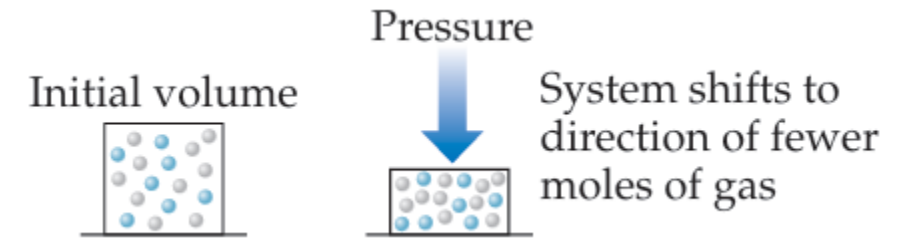
If a substance is added to a system at equilibrium, the system reacts to consume some of the substance. If a substance is removed from a system, the system reacts to produce more of the substance.



Le Châtelier's Principle (Henri-Louis Le Châtelier (1850–1936))

- **Pressure:** changing the pressure by changing the volume

At constant temperature, reducing the volume of a gaseous equilibrium mixture causes the system to shift in the direction that reduces the number of moles of gas.

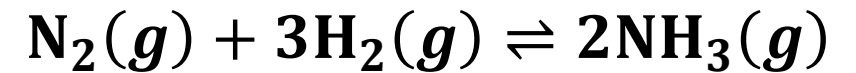
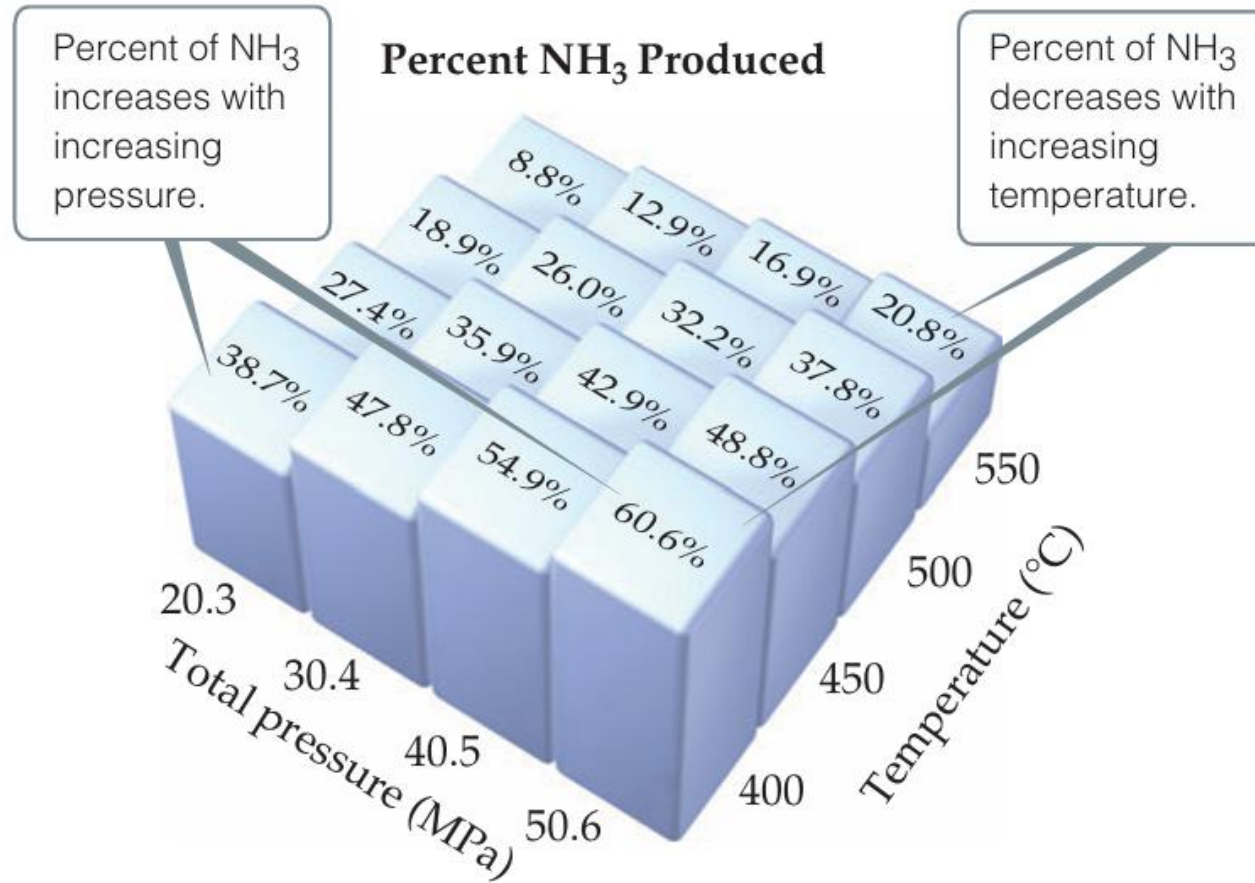


- **Temperature:**

If the temperature of a system at equilibrium is increased, the system reacts as if we added a reactant to an endothermic reaction or a product to an exothermic reaction. The equilibrium shifts in the direction that consumes the "excess reagent," namely, heat.



Le Châtelier's Principle (Henri-Louis Le Châtelier (1850–1936))

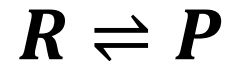


Q. At what combination of pressure and temperature should you run the reaction to maximize NH₃ yield?

Effect of temperature and pressure on NH₃ yield in the Haber process (Each mixture was produced by starting with a 3 : 1 molar mixture of H₂ and N₂)

Le Châtelier's Principle (Henri-Louis Le Châtelier (1850–1936))

Change in Reactant or Product Concentration

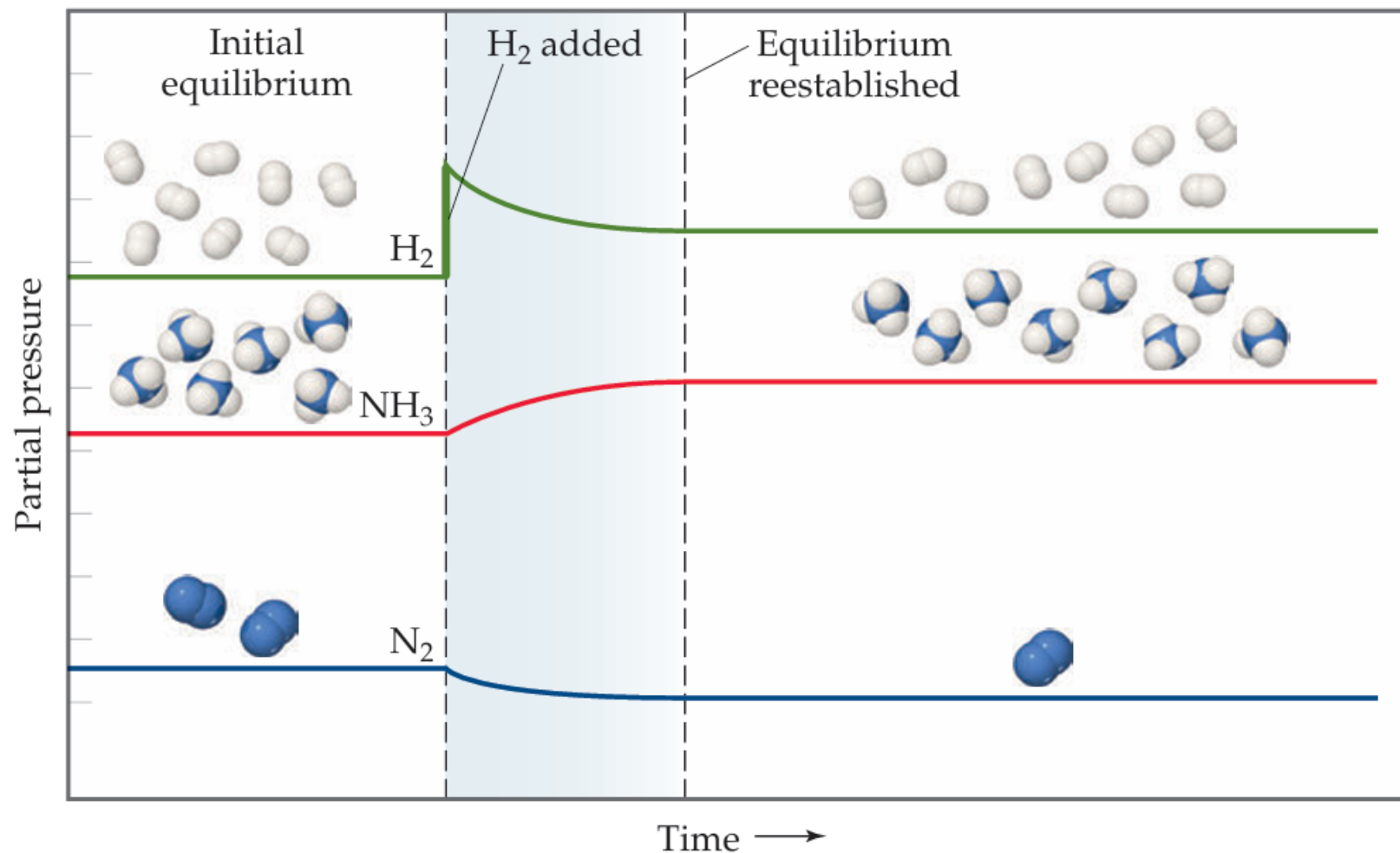
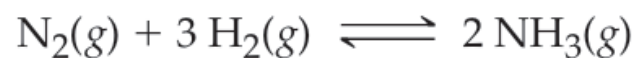


- ✓ If a chemical system is already at equilibrium and the concentration of any substance in the mixture is increased (either reactant or product), the system reacts to consume some of that substance.
- ✓ Conversely, if the concentration of a substance is decreased, the system reacts to produce some of that substance

Q. Do these types of changes affect the equilibrium constant?

Le Châtelier's Principle (Henri-Louis Le Châtelier (1850–1936))

Change in Reactant or Product Concentration



Effect of adding H_2 to an equilibrium mixture of N_2 , H_2 , and NH_3

Le Châtelier's Principle (Henri-Louis Le Châtelier (1850–1936))

Change in Reactant or Product Concentration

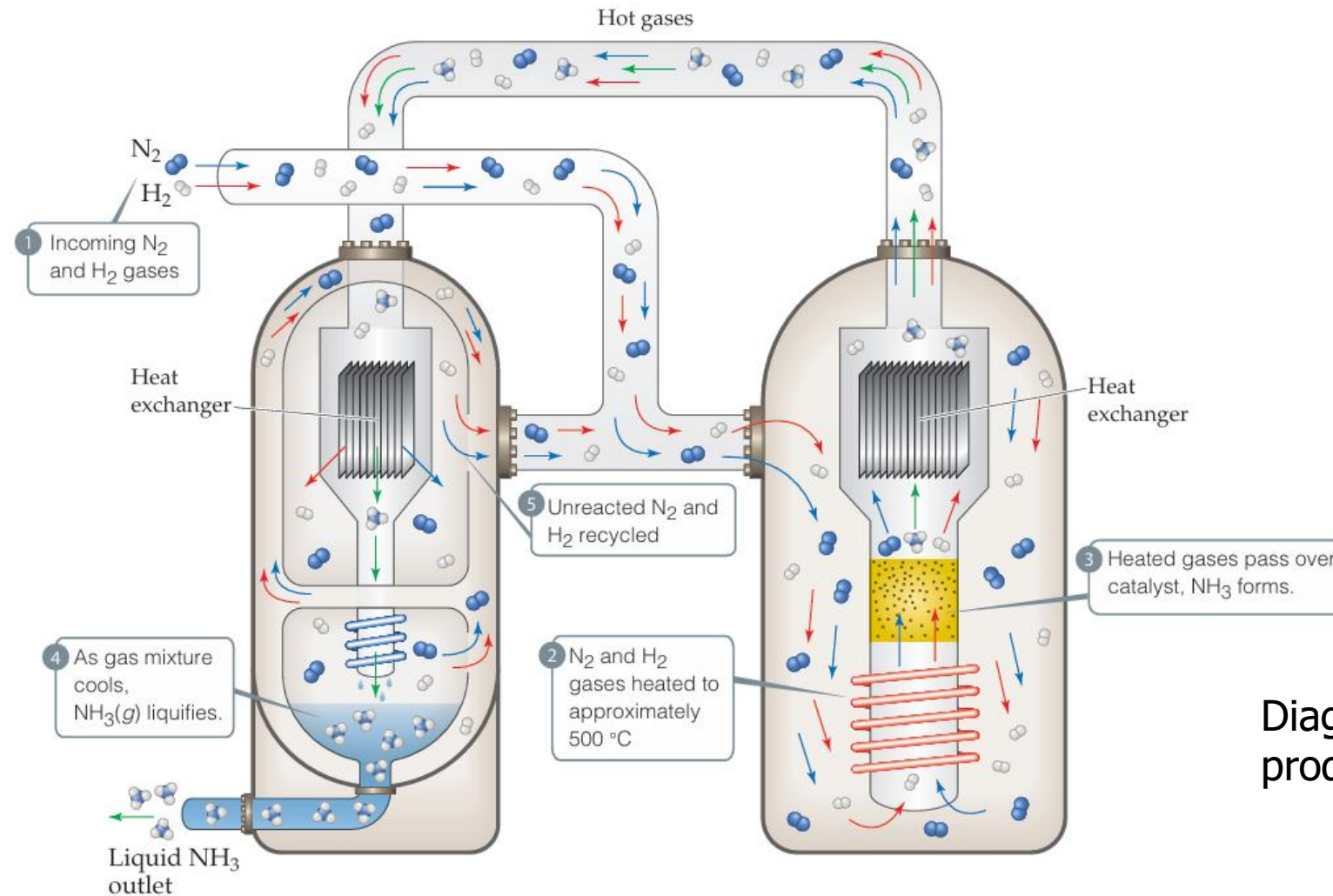


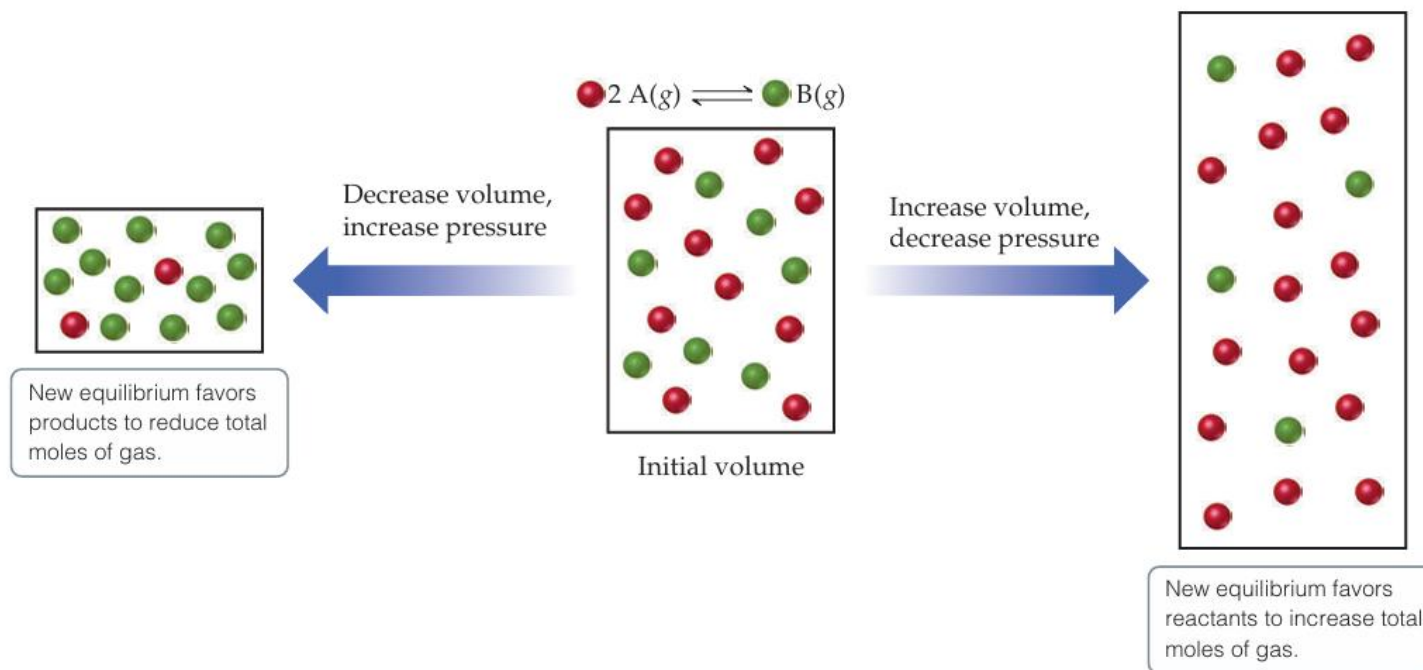
Diagram of the industrial production of ammonia

Le Châtelier's Principle (Henri-Louis Le Châtelier (1850–1936))

Effects of Volume and Pressure Changes

- ✓ At constant temperature, reducing the volume of a gaseous equilibrium mixture causes the system to shift in the direction that reduces the number of moles of gas. Increasing the volume causes a shift in the direction that produces more gas molecules.

Q. Do these types of changes affect the equilibrium constant?



- Changes in concentrations or partial pressures shift equilibria without changing the value of the equilibrium constant.
- In contrast, almost every equilibrium constant changes as the temperature changes.



Le Châtelier's Principle (Henri-Louis Le Châtelier (1850–1936))

Effect of Temperature Changes

- The relationship between K and temperature can be deduced from Le Châtelier's principle

Assumption:

- ✓ Endothermic (heat-absorbing) reaction, consider heat as a reactant.
- ✓ Exothermic (heat-releasing) reaction, we consider heat as a product.

Endothermic: Reactants + *heat* $\rightleftharpoons$ products

Exothermic: Reactants $\rightleftharpoons$ products + *heat*

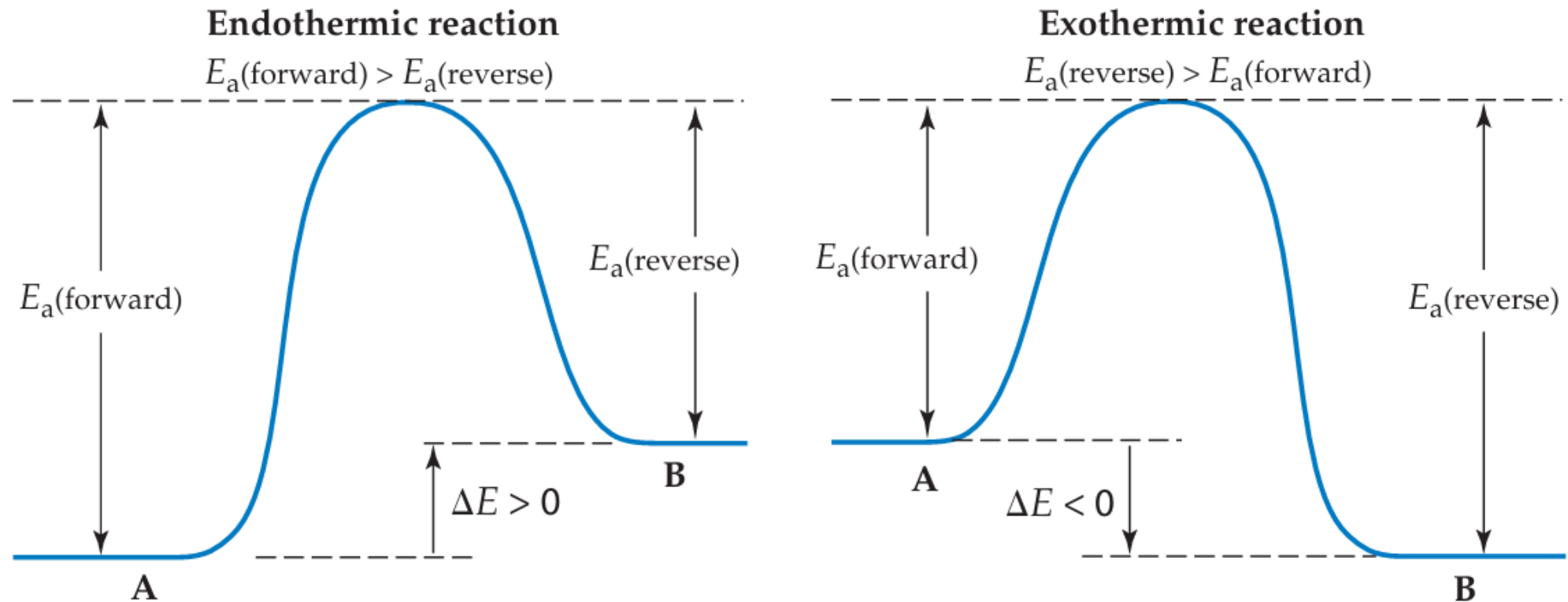
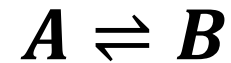
When the temperature of a system at equilibrium is increased, the system reacts as if we added a reactant to an endothermic reaction or a product to an exothermic reaction. The equilibrium shifts in the direction that consumes the excess reactant (or product), namely heat.

Endothermic: Increasing T results in higher K value

Exothermic: Increasing T results in lower K value

Le Châtelier's Principle (Henri-Louis Le Châtelier (1850–1936))

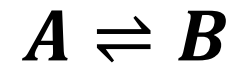
Effect of Temperature Changes



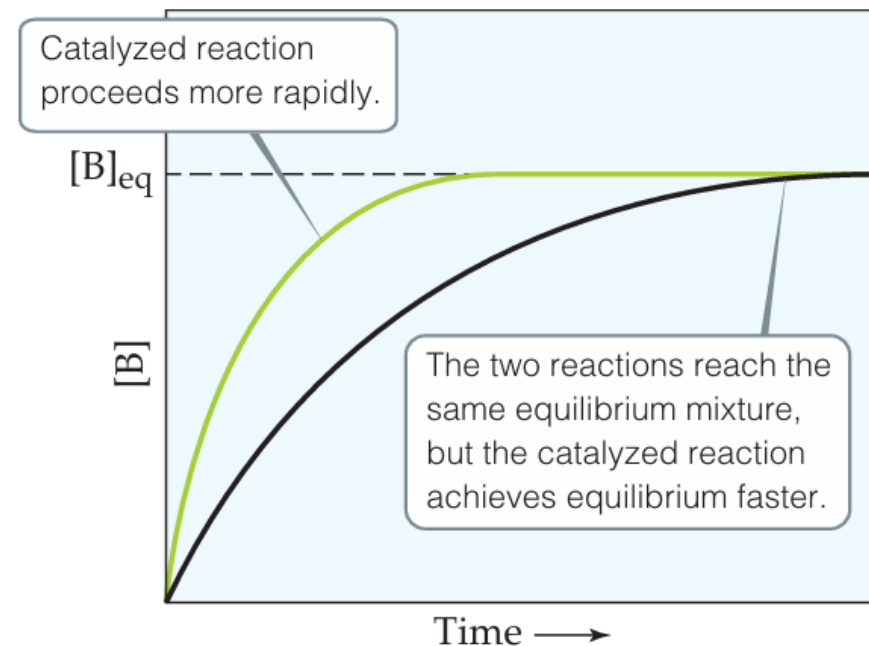
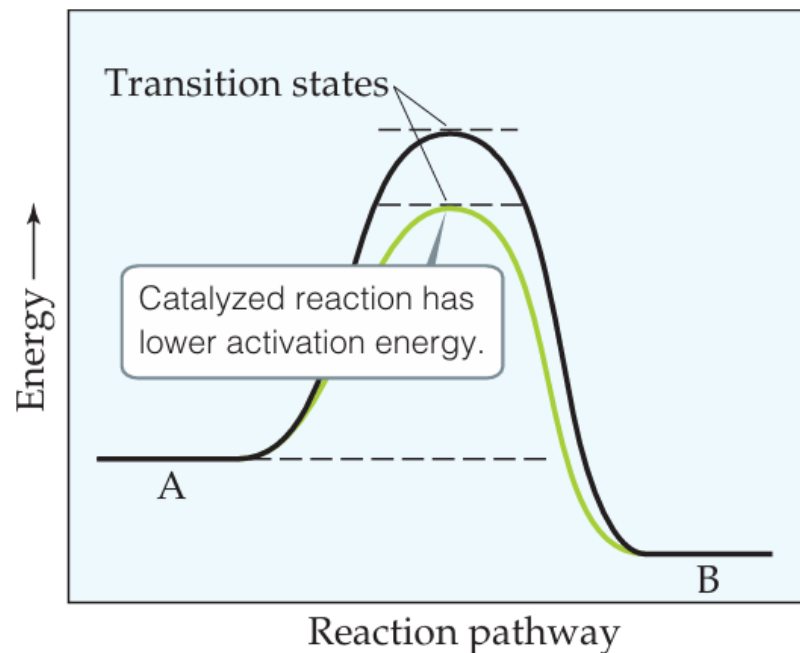
▲ **Figure 15.14** The energy profile for an endothermic reaction (left) and an exothermic reaction (right).

Le Châtelier's Principle (Henri-Louis Le Châtelier (1850–1936))

The Effect of Catalysts



Q. What happens if we add a catalyst to a chemical system that is at equilibrium?



An energy profile for the reaction $A \rightleftharpoons B$ (left), and the change in concentration of B as a function of time (right), with and without a catalyst. Green curves show the reaction with a catalyst; black curves show the reaction without a catalyst.

Summary Q.

- Q. At temperatures near 800 °C, steam passed over hot coke (a form of carbon obtained from coal) reacts to form CO and H₂:



The mixture of gases that results is an important industrial fuel called water gas.

- (a) At 800 °C the equilibrium constant for this reaction is $K_p = 14.3$. What are the equilibrium partial pressures of H₂O, CO, and H₂ in the equilibrium mixture at this temperature if we start with solid carbon and 0.100 mol of H₂O in a 1.00 L vessel?
- (b) What is the minimum amount of carbon required to achieve equilibrium under these conditions?
- (c) What is the total pressure in the vessel at equilibrium?
- (d) At 25 °C the value of K_p for this reaction is 1.7×10^{-23} . Is the reaction exothermic or endothermic?
- (e) To produce the maximum amount of CO and H₂ at equilibrium, should the pressure of the system be increased or decreased?

PP. 477-478

By now you should be able to:

- ✓ Explain what is meant by chemical equilibrium and how it relates to reaction rates.
- ✓ Write the equilibrium expression for any reaction.
- ✓ Convert K_c to K_p and vice versa.
- ✓ Relate the magnitude of an equilibrium constant to the relative amounts of reactants and products present in an equilibrium mixture.
- ✓ Manipulate the equilibrium constant to reflect changes in the chemical equation.
- ✓ Write the equilibrium expression for a heterogeneous reaction.
- ✓ Calculate an equilibrium constant from concentration measurements.
- ✓ Predict the direction of a reaction given the equilibrium constant and the concentrations of reactants and products.
- ✓ Calculate equilibrium concentrations given the equilibrium constant and all but one equilibrium concentration.
- ✓ Calculate equilibrium concentrations, given the equilibrium constant and the starting concentrations.
- ✓ Use Le Châtelier's principle to predict how changing the concentrations, volume, or temperature of a system at equilibrium affects the equilibrium position.

Key Equations

- $$K_c = \frac{[D]^d[E]^e}{[A]^a[B]^b}$$

[15.8]

The equilibrium expression for a general reaction of the type $aA + bB \rightleftharpoons dD + eE$, the concentrations are equilibrium concentrations only
- $$K_p = \frac{(P_D)^d(P_E)^e}{(P_A)^a(P_B)^b}$$

[15.11]

The equilibrium expression in terms of equilibrium partial pressures
- $$K_p = K_c(RT)^{\Delta n}$$

[15.15]

Relating the equilibrium constant based on pressures to the equilibrium constant based on concentrations
- $$Q_c = \frac{[D]^d[E]^e}{[A]^a[B]^b}$$

[15.24]

The reaction quotient. The concentrations are for any time during a reaction. If the concentrations are equilibrium concentrations, then $Q_c = K_c$.

Chapter 16

Acid–Base Equilibria

- ▶ Acid–Base Equilibria
- ▶ The Autoionization of Water
- ▶ The pH Scale
- ▶ Strong Acids and Bases
- ▶ Weak Acids
- ▶ Weak Bases
- ▶ Acid–Base Properties of Salt Solutions
- ▶ Acid–Base Behavior and Chemical Structure